

HAO YANG

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Actively seeking intern and full-time opportunities in Software Development starting May 2026.

EDUCATION

New York University

Brooklyn, NY

M.S. in Computer Engineering

Sep 2024 – May 2026

- Coursework: Machine Learning, High Performance Machine Learning, Deep Learning, Operating Systems, Cloud Computing, DevOps, Internet Architecture & Protocols, Java Programming

University of Washington

Seattle, WA

B.S. in Computer Science & Software Engineering; B.S. in Math, Visualization

Mar 2021 – Dec 2023

- Coursework: Data Structures & Algorithms, C/C++, Operating Systems, Databases, AI, Cybersecurity, Linear Algebra, Statistics, Hardware Organization, Networking

TECHNICAL SKILLS

- **Languages:** C/C++, Python, Java, Go, SQL, JavaScript, HTML/CSS, C#, ARM Assembly, YAML
- **Web/DevOps:** AWS, GCP, Docker, Kubernetes, REST, Redis, Git, PostgreSQL, MySQL, Maven, JWT
- **Frameworks:** FastAPI, Flask, Django, Spring Boot, React, Vue3, Unity3D
- **Machine Learning:** PyTorch, TensorFlow, Keras, NumPy, Pandas, MLflow
- **Deep Learning Systems:** CUDA, cuDNN, TorchScript, GPU Profiling, Model Inference Optimization
- **AI-Assisted Development:** Claude Code (agentic/vibe coding for full-stack feature development)

EXPERIENCE

Interview Bot

Seattle, WA

Software Development Engineer Intern

Sep 2023 – Dec 2023

- Containerized and migrated backend services and PostgreSQL from **AWS EC2** to **Google Kubernetes Engine (GKE)**, reducing deployment time by **60%**.
- Built **CI/CD pipelines** using GitHub Actions to automate build, test, and deployments across 3 microservices.
- Implemented **Kubernetes deployments**, ReplicaSets, HPA, load balancing, and configured IAM, VPC, Helm charts, and rollout strategies using **kubectl** and GCloud CLI.

PROJECTS

AIFriends — AI Companion Platform

Jan 2026 - Present

Python, Django, Vue 3, LangChain, LangGraph, LanceDB, Claude Code

- Architected a full-stack AI companion platform with a **Django REST Framework** backend and **Vue 3** SPA frontend, using **JWT** tokens (2h access / 7d refresh, rotated & blacklisted) with silent refresh across REST and **Server-Sent Events (SSE)**.
- Designed an **agentic RAG system** in **LangGraph** invoking a **LanceDB** vector-search tool via a self-authored **LangChain** adapter for a custom OpenAI-compatible LLM.
- Implemented **hierarchical memory**: the last 10 exchanges are periodically summarized into a long-term memory.
- Replaced the blocking, wait-for-full-response chat flow with token-by-token **SSE** streaming, surfacing the first token immediately instead of after full generation.

Smart Photo Album – AI Natural Language Search

Oct 2025 - Dec 2025

AWS, S3, Amazon Lex, OpenSearch, CloudFormation, Rekognition

- Engineered a serverless photo app on **AWS Lambda**, **API Gateway**, and **S3**: photos upload via presigned PUT URLs and are auto-indexed by an S3-triggered Lambda; **Amazon Lex** resolves search queries into keywords matched against **OpenSearch**.
- Built a pipeline merging **Amazon Rekognition** auto-labels (top **15** labels, $\geq 70\%$ confidence) with custom S3-metadata labels, deduplicated and indexed into **Amazon OpenSearch**.
- Defined the full stack (**API Gateway**, S3-to-Lambda triggers, least-privilege IAM, 3 Lambdas) as **CloudFormation** IaC via **AWS CodeBuild**.

Dining Concierge Chatbot

Sep 2025 - Oct 2025

AWS, Amazon Lex, API Gateway, DynamoDB, OpenSearch, Amazon SES

- Built a serverless dining chatbot where **Amazon Lex** handles multi-turn slot filling (location, cuisine, date/time, party, email) and a Lambda hook enforces booking rules (future dates, 20-person cap, valid email) before queuing to **SQS**.
- Designed the **REST API** with **API Gateway** and **AWS Lambda** as a Lex proxy, pinning session IDs to fix a slot-reset bug.
- Scraped and stored **1,600+** Manhattan restaurants across **8 cuisines** via the **Yelp API** into **DynamoDB** (full records) and **OpenSearch** (id + cuisine index).
- Automated recommendation delivery via a scheduled Lambda that queries up to **100** OpenSearch matches, randomly samples **5**, joins DynamoDB details, and emails picks via **Amazon SES**.